

Parametric vs Non-parametric

Parametric

- (+) More powerful when the assumption met.
- (+) More efficient
- (+) Provide the estimates of population parameters
- (-) Sensitive to assumption
- (-) Limited flexibility
- (-) May not capture complex relationship

Non-param

- (+) Robust to outliers.
- (-) Widely applicable
- (+) Easy to implement
- (-) May require larger sample sizes.
- (-) Less information about the population

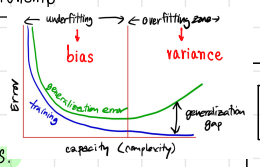
Common Assumption

Parametric

- Normality
- Homogeneity of variance
- Independence

Non-Parametric

- Random Sampling
- Homogeneity of Measurement
- Independence



Bias - Variance

$\hat{Y} = y_0 + \epsilon \approx f(X)$ [Random Variable]

$MSE(\hat{f}) = E[(f(X) - Y)^2]$

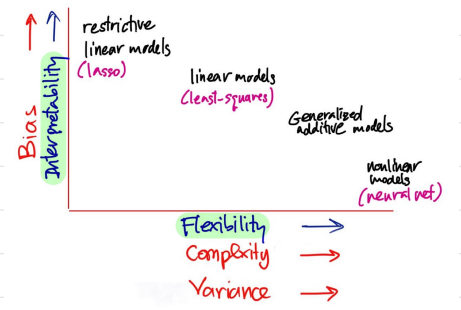
$= Var(f) + [Bias(f(X))]^2 + \sigma^2$

① $\epsilon \sim N(0, \sigma^2)$
② independently

$f(X) - y_0$

High Flexibility → bias ↓, variance ↑ = High Accuracy
Low Flexibility → bias ↑, variance ↓ = High Interpretability

variance = complexity.



Linear	Regression	↑ Parametric
Logistics	} Classification	
LDA		
QDA		
Naive Bayes		
KNN	} Non-param.	Non-param ↓ ?
Neural Network		

Cross-validation

→ เทคนิคในการแบ่งข้อมูล

Data → train, validation, testing.

the performance on validation set is used to estimate the performance when testing the model (deployment).

* เพื่อให้ภาพรวมกัน*

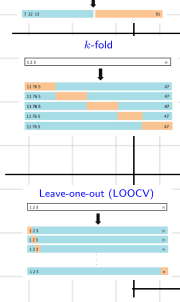
สมมติว่า $\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_d x^d$

Obj $f^* : \min_{\theta} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^d \theta_i^2 = J_{train}(\theta; \lambda, d)$

- ① objective เรามี 3 set of parameter θ, λ, d
- ② ในการเลือกโมเดลหนึ่ง เรา fix λ, d ไว้ก่อนหา θ
↳ โดยให้ J_{train} ต่ำสุด
- ③ เมื่อ J_{train} ต่ำสุด เราหาค่าของ J_{cv} สำหรับทุก (λ, d)
- ④ The best model คือ J_{cv} ที่ต่ำที่สุด ส่วน J_{test} ไว้ Report สายๆ

Split to two group

50% train
50% validate.



k-fold → randomly split data into k fold, leave 1 fold for validation.

- repeat k times where each time a different fold is regarded as validation set and compute MSE_i ; $i = 1, 2, 3, \dots, k$
- the test error rate is estimated by averaging the k MSE's.

LOOCV = k-fold, but with $k=m$ where m is the total number of sample or the total number of batches.

Properties	Hold out	k-fold	LOOCV
Time computational	low	feasible	high
# Samples	Ratio	$(k-1)m/k : m/k$	$m-1 : m$
Stability	Highly unstable	Intermediate	Stable

$\#train : \#valid$

Validation MSE ใกล้เคียงกัน

$L(\alpha) = -\frac{m}{2} \log 2\pi \sigma^2 - \frac{1}{2} m$

if # samples is large, AIC can be the best criterion for choosing a good balance model.

AICc(α) is the corrected version of AIC for small m; AICc(α) penalizes more on complexity (quadratic in m).

When # samples become large AICc → AIC

Model Selection Score = loss function (model quality) + model complexity.

$AIC(\alpha) = 2L(\alpha) + 2n$

$AICc(\alpha) = 2L(\alpha) + 2n + \frac{2n(n-1)}{m-n-1}$

$BIC(\alpha) = 2L(\alpha) + n \log m$

$FPE(\alpha) = V(\alpha) \left(\frac{1+n/m}{1-n/m} \right)$

$C_p(\alpha) = \frac{1}{m} (RSS(\alpha) + n\hat{\sigma}^2)^2$

Adjusted $R^2 = 1 - \frac{RSS(\alpha)}{TSS(\alpha)} \frac{m-1}{m-n-1}$

ยิ่งน้อยยิ่งดี
log-likelihood (L) นกนคือ
ยิ่งมากยิ่งดี

สำหรับ AIC, BIC (lower is better)

$AIC(\alpha_1) < AIC(\alpha_2) \leftrightarrow L(\alpha_1) - L(\alpha_2) > 2(n_2 - n_1)$

$BIC(\alpha_1) < BIC(\alpha_2) \leftrightarrow L(\alpha_1) - L(\alpha_2) > (n_2 - n_1) \log m$

ตัวอย่าง

สมมติ	A	B
มีตัวแปร (n)	2	11
Log-likelihood (L)	-500	-480
จำนวน	100	

$L(\alpha_1) - L(\alpha_2) = -20 > 2(11 - 2) = 18$ ✗
 $\therefore AIC(\alpha_1) \geq AIC(\alpha_2) \Rightarrow$ เลือก Model B ✗
 $L(\alpha_1) - L(\alpha_2) = -20 > (-9) \log(100)$ ✓
 $\therefore BIC(\alpha_1) < BIC(\alpha_2) \Rightarrow$ เลือก Model A ✗

Data set

- train set → ② J_{train}
- validation set → ③ J_{cv}
- test set → ④ J_{test}

Regularization Hyperparameter.

ถ้า coeff ใหญ่ → correlate ใหญ่
p-value ต่ำ = strong correlate

Linear Regression

Best Linear Unbiased Estimator (BLUE)

$\nabla_{\beta} \|y - X\beta\|_2^2 = 2X^T(y - X\beta) = 0$

$\beta_{LS} = (X^T X)^{-1} X^T y$

Convexity.
 $X^T X \geq 0 \rightarrow$ Positive definite matrix
↳ การหาค่า $(X^T X)^{-1}$ ได้
↳ หรือ β_{LS} ได้เฉพาะ

nullity ≠ 0

- ① หา Inverse Matrix ไม่ได้
- ② X_i, X_j บางคู่เป็น dependent กัน
→ แก้โดย Ridge, Feature Selection
PCA มาลดปัญหานี้

if $X \in \mathbb{R}^{m \times n}$ with $m \geq n$ is full column rank,

if $m=n$, then X is invertible, $(X^T X)^{-1} X^T = X^{-1}$ (linalg.inv()).

if $m > n$, then X is non-invertible, but

→ if nullity $(X^T X) = 0 \Rightarrow X^T X \geq 0$ Then, $(X^T X)^{-1} X^T$ can be used as the left inverse of X with pseudo inversion (linalg.pinv())

$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$; $\bar{y} = \frac{1}{n} \sum y_i$

Rejection Region: $\alpha_{scale} = -1.65 \frac{s}{\sqrt{n}} + \mu_0$

(Reject if $\bar{x} < \alpha_{scale}$ or $\frac{\bar{x} - \mu_0}{s/\sqrt{n}} < -1.65$)

Lower tailed

↑ Zero model params
Well encode sparse impact features.
(Label input features)
Enhancing model complexity (low runtime).

p-value < α ⇒ Reject H_0 with $(1-\alpha)100\%$.

Adjusted R^2 increases, if the added predictor variables decrease RSS enough to compensate for the increase in n.

All classifiers are a discriminative model. → A classifier

- generative LDA, QDA, Bayes
- discriminative SVM, Tree-Base, NN, Logistics.

Robust Regression

- Outliers: data points with extremely diverging measurements. 99% y (measurement)
- High leverages: data points with different mapping between input and output prediction. Usually, the problem lies in the input features. 99% X (input feature)

H-matrix $\rightarrow H = X^{Test} (X^{Train})^{-1} X^{Train}$, $\hat{y}^{Test} = X^{Test} \beta_{LS} = H y^{Train}$

$\hat{y}_{jj} = h_{jj} y_{jj} + \sum_{i \neq j} h_{ji} y_i^{train} = \sum_i h_{ji} y_i$; $\forall j$

The j^{th} data pair (x_j^i, y_j) is an leverage, if $h_{jj} > \frac{2n}{m}$

- Properties
- Symmetry: $h_{ij} = h_{ji}$
 - Bound: $h_{jj} \in [-0.5, 0.5]$, $h_{ij} \in [0, 1]$
 - Orthogonal: ถ้า $h_{jj} = 0$ แปลว่าข้อมูลสองจุดนี้ ไม่ส่งผลต่อกันเลย (in feature space) การเปลี่ยนค่า y_j จะไม่ทำให้ $\hat{y}_j$ เปลี่ยน
- ถ้าเราเปลี่ยน y_i เป็น y'_i แล้ว $\hat{y}_j$ จะเปลี่ยนเป็น $\hat{y}'_j = \hat{y}_j + h_{ji} (y'_i - y_i)$; $\forall i, j$

Classification

		Predicted samples		
		Predicted Positive	Predicted Negative	
Actual truth	Actual Positive (P)	True Positive (TP)	False Negative (FN) $FN = P - TP$	Recall, Sensitivity True Positive Rate (TPR) $\frac{TP}{P}$ Type II error, False Negative Rate (FNR) $\frac{FN}{P} = 1 - TPR$
	Actual Negative (N)	False Positive (FP) Type I	True Negative (TN) $TN = N - FP$	Type I error, False Positive Rate (FPR) $\frac{FP}{N}$ Specificity, Selectivity, True Negative Rate (TNR) $\frac{TN}{N} = 1 - FPR$
		Precision $\frac{TP}{TP + FP}$	Accuracy $\frac{TP + TN}{P + N}$	

$F1 = \frac{2 \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$

Prevalence: $= \frac{\text{total positive}}{\text{total population}} = \frac{P}{P+N}$

False discovery rate (FDR, FPR) $= \frac{\text{incorrectly predicted positive}}{\text{predicted positive}} = \frac{FP}{FP+TP}$

Positive predictive value (PPV): $= \frac{\text{correctly predicted positive}}{\text{predicted positive}} = \frac{TP}{TP+FP}$

$= 1 - \text{FDR (or precision)}$

Sensitivity-specificity trade-off
Precision-recall trade-off

confusion matrix of C-classes.

Actual \ Predicted	G1	G2	G3	G4	G5	sum
G1	9	2	2	4	2	19
G2	1	10	2	2	5	20
G3	2	5	14	1	5	27
G4	1	5	2	8	3	19
G5	1	2	2	2	8	15
sum	14	24	22	17	33	100

$Acc = \frac{\sum TP_c}{\text{All predict}} = \frac{9+10+14+8+3}{100}$

Precision $= \frac{TP_c}{TP_c + FP_c} \rightarrow Pre_{G3} = \frac{14}{14+8} (\#G3)$

Recall $= \frac{TP_c}{TP_c + FN_c} \rightarrow Re_{G3} = \frac{14}{14+13} (\#G3)$

Overall the performance

- Macro $\rightarrow$ Precision $= \frac{1}{C} \sum_c Precision_c$; Recall $= \frac{1}{C} \sum_c Recall_c$
F1-score $= 2 \cdot \frac{Prec \cdot Recall}{Prec + Recall}$
ข้อเสีย small C $\rightarrow$ make the classifier look "worse" due to low performance in an unimportant class.
large C $\rightarrow$ disguise poor performance in the critical minority class when #C is large.
In this case, the contribution of each class is diluted.

②. Micro $\rightarrow$ Precision $= \frac{\sum_c TP_c}{\sum_c (TP_c + FP_c)} (\#G_c)$; Recall $= \frac{\sum_c TP_c}{\sum_c (TP_c + FN_c)}$

Accuracy = F1-score = Precision = Recall

จำนวน data รวม = N.

③. Weighted $\rightarrow$ Precision $= \sum_c w_c Precision_c$; Recall $= \sum_c w_c Recall_c$

F1-score $= 2 \cdot \frac{Prec \cdot Recall}{Prec + Recall}$

Proof Decision Boundary for 2D Gaussians

ให้ $X|Y=c \sim \text{Normal}(\mu_c, \Sigma_c)$ โดย $\mu_c = \begin{bmatrix} \mu_{c1} \\ \mu_{c2} \end{bmatrix}$ และ $\Sigma_c = \begin{bmatrix} \sigma_{c1}^2 & \sigma_{c1}\sigma_{c2}\rho \\ \sigma_{c1}\sigma_{c2}\rho & \sigma_{c2}^2 \end{bmatrix}$

ทั้ง $\mu_{ic} = \frac{\text{ค่า X ทั้งหมดตาม features i, Class c}}{\text{จำนวนค่า X features i, Class c}}$ และ σ_{ij}^2, ρ ที่คำนวณตามปกติ โดยแยกตาม c

99% $g_c(x) = \log[P(X|Y=c)] + \log \pi_c$

$= \log \left\{ \frac{1}{(2\pi)^n |\Sigma_c|} \exp \left[-\frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c) \right] \right\} + \log \pi_c$

$= -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log(\det \Sigma_c) - \frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c) + \log \pi_c$

$= -\frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c) - \frac{n}{2} \log(2\pi) + \log \pi_c$

$g_c(x) = -\frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c) - \frac{n}{2} \log(\det \Sigma_c) - \frac{n}{2} \log(2\pi) + \log \pi_c$

$= -\frac{1}{2} (x^T \Sigma_c^{-1} x + 2 \mu_c^T \Sigma_c^{-1} x + \mu_c^T \Sigma_c^{-1} \mu_c) - \frac{1}{2} \log(\det \Sigma_c) - \frac{n}{2} \log(2\pi) + \log \pi_c$

$g_c(x) = -\frac{1}{2} x^T \Sigma_c^{-1} x - \mu_c^T \Sigma_c^{-1} x - \frac{1}{2} \mu_c^T \Sigma_c^{-1} \mu_c - \frac{1}{2} \log(\det \Sigma_c) - \frac{n}{2} \log(2\pi) + \log \pi_c$

quadratic $A = \frac{1}{2} \Sigma_c^{-1}$, linear term $B = -\mu_c^T \Sigma_c^{-1}$, constant C

Decision Rule: $g_1(x) = g_2(x)$

$x^T (A_1 - A_2) x + (B_1 - B_2) x + (C_1 - C_2) = 0$

ถ้า $\Sigma_1 = \Sigma_2 = \dots = \Sigma_c = \Sigma$

Class ทุก Class มี covariance matrix เท่ากัน $\rightarrow$ แทนที่จะใช้การตัดสินใจ $\rightarrow$ LDA.

	AIC	BIC	AICc
Target	High Prediction	Find the true model	หลีกเลี่ยง AIC เมื่อ m น้อยๆ
Risk	เสี่ยง Overfit	เสี่ยง Underfit	ดีสำหรับ m น้อยๆ
เมื่อ m ใหญ่ๆ	-	ค้นหา true model (consistent)	ค่าเท่ากับ AIC

Training errors always decrease as model complexity increase.

Generalization error initially decreases as model picks up relevant features of data.

However, if the model complexity exceeds a certain degree, the generalization error can rise up again — this is when we observe overfitting.

จำนวน $\hat{\mu}_c = \{\hat{\mu}_{c1}, \hat{\mu}_{c2}, \hat{\mu}_{cn}\}$

①. $\hat{\mu}_c = [\hat{\mu}_{c1}, \hat{\mu}_{c2}, \dots, \hat{\mu}_{cn}]^T$

②. $\hat{\Sigma}_c = \begin{bmatrix} \sigma_{c1}^2 & \sigma_{c1}\sigma_{c2}\rho & \dots \\ \sigma_{c1}\sigma_{c2}\rho & \sigma_{c2}^2 & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$

③. $\hat{\pi}_c$ (bias term)

เมื่อ C เพิ่มขึ้น: ทำให้ # parameters ใน QDA >> LDA ส่งผลให้ Lead ไปสู่การ "Overfit" ได้ง่าย.

หากเลือก: C-1 เพราะหากเลือก C-1 class สัก class 1 - $\sum_{j=1}^{C-1} \hat{\mu}_j$

Expo $\rightarrow P(x_j|Y=c) = \lambda_{cj} e^{-\lambda_{cj} x_j}$

Discriminant f^3 : $g_c(x) = \ln[P(Y=c)] + \sum_{j=1}^n \ln[P(x_j|Y=c)]$

conditional independent probabilities.

Poisson $\rightarrow P(x_j|Y=c) = \frac{\lambda_{cj}^{x_j}}{(x_j)!} e^{-\lambda_{cj}}$

Why Validation? $\rightarrow$ Estimate Generalization Error

$\rightarrow$ Model Selection / Hyperparameter Tuning

$\rightarrow$ Detect Overfitting.

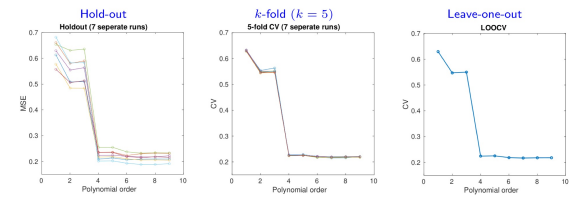
Residual Standard Error

Residual Sum Square

$RSE = \sqrt{\frac{\sum_j (y_j - \hat{y}_j)^2}{m-p}}$

$\rightarrow Y = \beta_0 + \beta_1 X \Rightarrow p=2$

$\rightarrow$ ถ้าหารแล้ว m จะ underestimate var



- Results:
- Hold-out has high variation since it depends on random splitting.
 - k-fold (k=5) has less variation because MSE is averaged over 5 folds.
 - LOOCV requires running $N=500$ loops (high computational cost), but the MSE's are highly correlated.

$RSS(u) := \sum_{j=1}^m (y_j - \hat{y}_j)^2$; each $\hat{y}_j$ is estimated value from the model and parameter defined in α

For AIC, AICc, BIC, FPE, C_p , the one with lower score is the better choice.

For Adjusted R^2 , the one with higher score is the better choice.

ทำ Validation.

Derive input feature.

RSS, R^2 ...

อันไหนเป็น outlier.

p-value หรือ
↳ คำนวณ (ทำ)

คำนวณ $\hat{H}$ ไม่ยาก.

Robust → H-matrix

คุณใหม่.

ทำไม β ไม่ใช่?

~~Cook distance~~

ทำไม Cook distance.

วิเคราะห์ AIC, BIC, AICc
ทำไม?

ทำ Validation.

โวลุ่มนี้เส้น Curve Performance

High Var, High Bias

Linear Model ที่คือ
ที่ Overfit

ที่ Complexity ที่น้อยเกินไป,

ถ้า linear - |coeff| น้อย → ไม่ค่อย correlate.

Output performance. ใน คำนวณ R^2 ใหม่.

ไม่ออกพวก Weighted LS } Robust.
Iterative พวกนี้